From Numerical Discrepancy to Task-Level Risk: Predicting the Effects of GPU Kernel Replacement

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Abstract

GPU kernel replacement is commonly screened by comparing kernel outputs against a numerical tolerance, but local agreement does not guarantee task-level agreement. We conduct a comparative study of twelve estimators trained on 6,928 grouped perturbation observations and transferred to 2,739 real replacements. Evaluation uses shared balanced draws over 21 risk-budget pairs. Mean F1 ranges from 0.4867 for a prevalence baseline to 0.8484 for weighted nearest neighbors; a grouped neural network and a jointly fitted random forest score 0.8460 and 0.8411. The leading models change across budgets, and their average scores are close relative to repeat-level sampling variation. The results characterize trade-offs among model assumptions rather than identify a universally best estimator.

1 Introduction

Modern model-serving systems rely on highly optimized GPU kernels. Replacing a reference implementation with a faster kernel can improve throughput, but changes in reduction order, fused arithmetic, or intermediate precision can create floating-point discrepancies. Numerical analysis explains how such errors accumulate through computation graphs [1, 2, 3]. Recent inference studies also show that execution settings and low-precision arithmetic can alter model outputs [4, 5].

Kernel-generation benchmarks make local correctness checks practical by comparing candidate-kernel outputs with a reference under a numerical tolerance [6, 7]. A passing check establishes that the compared tensors are close on the probe inputs. It does not establish that a replacement preserves the model's final class, answer, transcript, or correctness decision. The same numerical discrepancy can have different task-level effects depending on the affected operator and inference context.

We study whether controlled perturbations can bridge local discrepancy measurements and task-level behavior. Perturbation measurements vary discrepancy magnitude while preserving the inference context. Real replacements provide a different discrepancy-generating mechanism for evaluation. We freeze each estimator before scoring replacement outcomes. The first endpoint is an *output change*: the task-level discrete output differs from the reference. The second is a *negative flip*: a request that the reference execution answers correctly becomes incorrect under the replacement. A negative flip is a subset of output changes, but it also depends on the reference model's correctness.

We follow the supplied Kernel-Correctness study's random-forest configuration, 13-role taxonomy, and balanced replacement protocol [8]. We compare twelve fixed estimators spanning prevalence, monotone regression, linear and spline models, kernel regression, nearest neighbors, a neural network, single trees, and tree ensembles. Replacement outcomes are not used to fit models or choose among them. Each estimator first produces predictions from replacement features. The same fixed balanced draw manifest then scores every estimator.

This course project makes three empirical contributions:

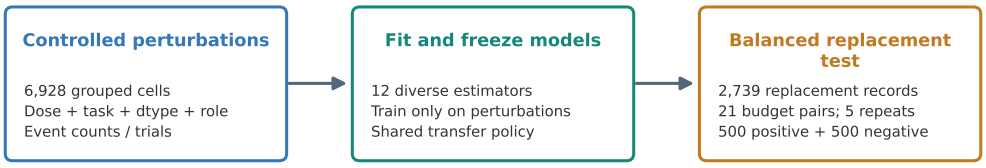


Figure 1: Paper-aligned workflow. Models are fit on grouped controlled-perturbation observations, frozen, and transferred to real replacements using the measured maximum discrepancy. Replacement labels are opened only after feature-only predictions have been saved.

- We compare twelve model families for two grouped-binomial task-risk outcomes under a frozen perturbation-to-replacement protocol.
- We characterize how aggregate scores, severity-specific recall, and budget sensitivity vary with estimator assumptions.
- All formal results use the same balanced replacement test draws, so comparisons share class allocation and sampling noise.

2 Related Work

Numerical variation in model inference. Floating-point behavior depends on arithmetic precision, evaluation order, and accumulated rounding error [1]. Accuracy-aware compilation and hardware emulation study how numerical deviations can be bounded or exposed before deployment [2, 3]. FlashAttention-3 uses low precision and asynchronous execution for speed [5], while Yuan et al. [4] show that execution settings can alter language-model outputs. The supplied Kernel-Correctness study connects controlled perturbations to measured replacement outcomes and specifies the role taxonomy and balanced evaluation protocol used here [8].

Kernel generation and local correctness. KernelBench evaluates generated GPU kernels for efficiency and local correctness [6]. TritonBench and MultiKernelBench extend evaluation to Triton operators and additional platforms [7, 9]. We use the local discrepancy measured in this stage as a feature for predicting task-level outcomes after replacement.

Risk estimation. Isotonic regression is suited to settings where estimated risk should not decrease as discrepancy grows. Pooled-adjacent-violators methods produce monotone, nonparametric curves [10]. Our comparison spans ridge and logistic regression, RBF support-vector regression, nearest-neighbor smoothing, a neural network, a shallow tree, random forests, ExtraTrees, histogram gradient boosting, and CatBoost [11, 12, 13]. These methods encode different assumptions about smoothness, locality, partitions, and ensemble construction. Work on preserving sample-level correctness across model updates provides a related correctness endpoint [14]. Our setting keeps model parameters fixed and changes the kernel implementation.

3 Methodology

Figure 1 summarizes the separation between controlled training data and real-replacement evaluation. We adopt the source study's perturbation-to-replacement protocol and balanced sampling design [8].

3.1 Task and grouped observations

Let $f_{\text{ref}}(x)$ be the final output of a reference model execution on request x, and let $f_{\text{alt}}(x)$ be the output after perturbing or replacing one kernel. Let $C(o, y) \in \{0, 1\}$ indicate correctness against the task target y. We define

$$Z_{\text{change}} = \mathbf{1}\{f_{\text{alt}}(x) \neq f_{\text{ref}}(x)\}, \qquad Z_{\text{flip}} = \mathbf{1}\{C(f_{\text{ref}}(x), y) = 1 \wedge C(f_{\text{alt}}(x), y) = 0\}. \quad (1)$$

The output comparison uses the task's discrete result: top-1 label for classification, selected answer for multiple-choice QA, parsed answer for GSM8K, extracted span for extractive QA, and decoded transcript for ASR. Negative flip uses the reference-correct-to-alternative-wrong transition.

The perturbation corpus is represented as grouped-binomial observations. A cell is identified by model, logical kernel, dtype, benchmark, and perturbation setting. For endpoint e, cell i contains $k_{i,e}$ events among n_i requests, with empirical rate $r_{i,e} = k_{i,e}/n_i$. We retain event and trial counts rather than expanding rates into pseudo-independent rows. The training snapshot has 6,928 cells and 1,299 complete contexts spanning 13 models, seven task scopes, and FP32, FP16, and BF16.

3.2 Inputs and discrepancy mapping

The main predictor uses four inputs: $\log_{10}(d)$, task scope, dtype, and the paper's 13-level operator role. The role taxonomy separates attention linear, attention arithmetic, attention PV, attention softmax, normalization, activation, convolution, pooling, MLP projection, generic projection, output head, logit scale, and residual scale. Model identity, candidate identity, and benchmark identity remain available for grouping and diagnostics but are not predictor features.

For a perturbation cell, d is the upper endpoint of its applied magnitude range, denoted `range_high`. For a real replacement, we use the maximum absolute difference between matched reference and optimized kernel outputs, denoted `max_abs_diff`. This is a max-to-max mapping that preserves the interpretation of the paper's upper-bound dose. An exact-zero replacement discrepancy receives predicted risk zero. A nonzero value outside role-specific support is evaluated at the nearest supported boundary, with fallback from task–dtype–role to task scope and then to pooled training support.

3.3 Compared estimators

We compare twelve estimators with fixed settings. The prevalence baseline predicts request-weighted training prevalence. Dose-only isotonic regression uses weighted pooled adjacent violators and smoothed rate $(s + 0.5)/(n + 1)$. Ridge regression uses $\alpha = 1$. The multinomial softmax logistic model uses a spline basis of log-dose with task, dtype, and role interactions. RBF support-vector regression uses $C = 1$ and $\epsilon = 0.02$. The nearest-neighbor estimator averages the 25 closest cosine-distance observations, weighted by trial count and inverse distance. The grouped MLP has two hidden layers (64 and 32 units) and minimizes trial-weighted grouped categorical cross-entropy for the nested outcomes. A single decision tree is restricted to depth 2. ExtraTrees uses 120 randomized trees, depth 8, and leaf size 6. Histogram gradient boosting uses 180 iterations and at most 15 leaves. CatBoost uses CPU MultiRMSE with 220 iterations, depth 4, learning rate 0.05, and L2 regularization 5.

The jointly fitted random forest contains 120 trees, each with maximum depth 8 and minimum leaf size 6; every available feature is considered at each split, and the fixed random seed is 20260905. It regresses both observed rates in a multi-output model. During fitting, the negative-flip target is scaled by two and restored to its original scale at prediction. The shared post-fit projection enforces $p_{\text{flip}} \leq p_{\text{change}}$. This specification provides a reproducible forest benchmark alongside the alternatives; it does not receive a separate tuning budget.

All learned predictions are clipped to $[0, 1]$ and projected to satisfy $p_{\text{flip}} \leq p_{\text{change}}$. To apply one consistent transfer policy, each context is evaluated on a 61-point geometric dose grid. We replace raw predictions by their nondecreasing upper envelope and use an upper-step lookup for the observed replacement discrepancy. This post-fit policy makes the monotonicity and support behavior explicit for every estimator.

3.4 Locked replacement evaluation

The replacement pool contains 2,739 unique records from the current packaged snapshot. We fit each estimator on all perturbation cells, save feature-only replacement predictions, and then open the replacement event counts for scoring. For each budget pair, a model flags a record when either predicted endpoint rate exceeds its corresponding budget. The observed joint-positive label is defined analogously from the two measured rates. The primary score uses 21 pairs drawn from budgets of 0.1%, 0.2%, 0.5%, 1%, 2%, and 5%, retaining pairs where the negative-flip budget is no larger than the output-change budget.

For every pair, we sample 500 positive and 500 negative records with replacement. Within each class and net-accuracy-drop severity band, contexts receive equal total weight. The severity bands are at most 0.5%, above 0.5% through 1%, and above 1%. Accuracy drop is the net correct-to-wrong count after subtracting wrong-to-correct transitions. We repeat each pair five times and reuse the draw manifest across all models, for 105,000 draw positions. The reported F1 values average across budget pairs and repeats. Their uncertainty summary is the mean of the 21 within-budget sample standard deviations across five repeats. Because draws overlap and are sampled with replacement, it describes resampling variability rather than independent-record uncertainty. Balanced scores measure screening discrimination and do not estimate natural replacement prevalence. We inherit master seed 9 from the reference protocol; the source study reports selecting it retrospectively from 128 candidates [8]. Our run does not search over seeds, so the five-repeat standard deviations do not include uncertainty from that earlier seed selection.

For a sampled record, let $y = 1$ denote an observed joint-positive replacement and $\hat{y} = 1$ denote a flag from the model's two predicted risks and the selected budget pair. Precision is $TP/(TP+FP)$, joint recall is $TP/(TP + FN)$, and F_1 is the harmonic mean of precision and recall. The accepted set consists of records with $\hat{y} = 0$. Its violation fraction is $FN/(FN + TN)$: among records the model would accept, this is the proportion that are actually positive. We report recall within the two net-accuracy-drop bands as a separate severity diagnostic. All measures are computed per draw, then summarized over repeats and budgets as specified above. Since every draw contains equal numbers of positives and negatives, the reported precision is specific to this balanced design.

4 Experiments

4.1 Experimental setup

The run uses the current normalized project snapshot and no new GPU execution. The perturbation input contains 6,928 cells and 1,299 contexts. The replacement feature table contains 2,739 records and the same 1,299 context identities. The implementation uses NumPy 2.1.3, pandas 2.2.3, and scikit-learn 1.6.1. Tree and CatBoost models run on CPU. Every model uses the same role mapping, dose mapping, support policy, and balanced draw manifest. Replacement labels are an external evaluation input rather than a source of model selection.

4.2 Main comparison

Table 1 reports the mean over the 21 budget pairs and five fixed repetitions. Joint recall is recall for the union positive class defined by the two budget limits. The two accuracy-drop columns measure recall on records whose net accuracy drop exceeds 0.5% and 1%, respectively. Accepted violation fraction is the fraction of records predicted safe that are nevertheless positive under the joint budget label.

Weighted k-nearest neighbors has the highest mean F1 (0.8484), followed by the grouped MLP (0.8460) and the jointly fitted random forest (0.8411). The respective mean within-budget repeat standard deviations are 0.0112, 0.0108, and 0.0115. These standard deviations describe variation across five repeated balanced draws within each budget, averaged over budgets; they are not confidence intervals for the difference between models. The score gaps are smaller than this resampling scale, so the aggregate ranking is descriptive and does not establish a reliable winner. The alternatives also differ in threshold behavior: softmax logistic has the lowest accepted violation fraction (0.1353), while the depth-2 tree has the highest recall in the $> 0.5\%$ and $> 1\%$ net-accuracy-drop

Table 1: Balanced replacement screening across twelve estimators. F1 is the mean across 21 budget pairs and five repeats, followed by the mean within-budget sample SD. All results use the shared balanced test draws.

Method	F1 ↑ (mean ± SD)	Precision ↑	Joint recall ↑	$> 0.5\%$ recall ↑	$> 1\%$ recall ↑	Violation ↓
Prevalence	0.4867 ± 0.0066	0.4097	0.6000	0.6388	0.6497	0.2879
Dose-only isotonic	0.7468 ± 0.0104	0.6712	0.8473	0.8521	0.8480	0.2118
Ridge regression	0.8075 ± 0.0113	0.7490	0.8801	0.8154	0.7105	0.1445
Softmax spline logistic	0.8338 ± 0.0116	0.7954	0.8784	0.8239	0.7267	**0.1353**
RBF support-vector regression	0.8020 ± 0.0103	0.7386	0.8827	0.8411	0.7624	0.1444
Weighted k-nearest neighbors	**0.8484** ± 0.0112	**0.8311**	0.8678	0.8042	0.6951	0.1373
Grouped MLP	0.8460 ± 0.0108	0.8261	0.8686	0.7977	0.6770	0.1381
Decision tree (depth 2)	0.7897 ± 0.0119	0.7860	0.8168	**0.8506**	**0.7905**	0.1886
Joint multi-output random forest	0.8411 ± 0.0115	0.8220	0.8634	0.8375	0.7638	0.1419
ExtraTrees	0.8384 ± 0.0113	0.8164	0.8638	0.8274	0.7453	0.1429
HistGradientBoosting	0.8384 ± 0.0117	0.8098	0.8710	0.8302	0.7433	0.1385
CatBoost	0.8346 ± 0.0119	0.8017	**0.8727**	0.8409	0.7626	0.1392

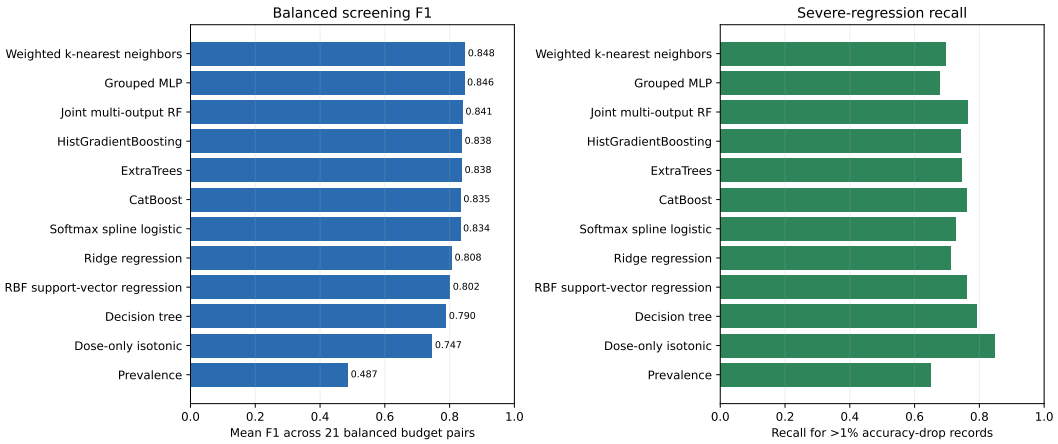


Figure 2: Balanced screening F1 and recall for >1% net-accuracy-drop records across all twelve fixed estimators. Both panels use the same paper-protocol balanced test draws.

subgroups (0.8506 and 0.7905). The highest mean F1 therefore does not imply the best performance for every operational objective.

The score range is wide across modeling assumptions. The prevalence baseline reaches 0.4867 mean F1, showing that a constant risk estimate does not adequately sort records for this balanced screening task. The dose-only monotone curve reaches 0.7468, a substantial increase over the constant baseline, while models that combine dose with context generally score between 0.79 and 0.85. This ordering is consistent with context adding useful information, but it is not a controlled feature ablation: contextual models also differ in functional form, regularization, and optimization. In particular, the current experiment does not isolate the marginal contribution of dtype or operator role while holding the estimator fixed.

Across the twelve models, no single inductive bias dominates all reported criteria. Local averaging performs well on mean F1 and precision; the neural network is close on mean F1; the spline logistic model has the lowest accepted violation fraction; and the shallow tree emphasizes recall in the larger net-accuracy-drop groups at the cost of its aggregate F1. CatBoost has the largest joint recall (0.8727), but its mean F1 (0.8346) is below the three leading methods. These patterns illustrate why threshold-dependent screening should be judged against a specified cost: missing a risky replacement, rejecting a safe one, and preserving severe-regression recall are different objectives.

4.3 Interpretation within the balanced test set

The formal test set intentionally gives equal positive and negative class counts for each budget pair. Therefore, F1 and precision describe a balanced screening task and should not be read as natural production prevalence or as calibrated deployment probabilities. Because all methods share the exact draw manifest, score differences are paired at the sampled-record level, although the reported

Table 2: F1 sensitivity across the 21 balanced budget pairs. Range endpoints and standard deviation are computed over pair-level means. First-place counts use the highest pair mean, with no ties in this run.

Estimator	F1 range	SD across budgets	First-place pairs
Prevalence	0.000–0.765	0.3530	0
Dose-only isotonic	0.703–0.800	0.0293	0
Ridge regression	0.765–0.864	0.0261	0
Softmax spline logistic	0.790–0.874	0.0271	1
RBF support-vector regression	0.770–0.870	0.0237	1
Weighted k-nearest neighbors	0.817–0.871	0.0174	12
Grouped MLP	0.800–0.896	0.0263	6
Decision tree (depth 2)	0.711–0.844	0.0439	0
Joint multi-output random forest	0.808–0.883	0.0229	1
ExtraTrees	0.802–0.874	0.0200	0
Histogram gradient boosting	0.786–0.873	0.0237	0
CatBoost	0.776–0.876	0.0305	0

summary does not itself provide a paired confidence interval. Balanced sampling also changes the effective class frequency, so accepted fractions and precision cannot directly estimate production workload or the fraction of real replacements that should be rejected. Those quantities require evaluation under the deployment distribution and an explicit operating threshold.

The columns in Table 1 also separate ranking from the policy implied by a fixed risk threshold. Weighted KNN combines the highest mean precision (0.8311) with joint recall of 0.8678, while CatBoost reaches higher joint recall (0.8727) at lower precision (0.8017). The depth-2 tree provides a more pronounced severity trade-off: it recalls 0.7905 of records with net accuracy drop above 1%, but its overall joint recall is 0.8168 and its accepted violation fraction is 0.1886. A team prioritizing broad risk coverage might prefer a different operating point from one limiting false alarms. Because each estimator emits two endpoint risks and the flag rule is an OR across budget exceedances, those preferences should ultimately be expressed through separate costs for output change, negative flip, and severe accuracy loss.

The violation statistic has a conditional denominator: it is the share of predicted-safe sampled records that are actually positive under the joint label. It does not count missed risky records as a fraction of all positives; that quantity is captured by one minus joint recall. For example, softmax logistic has the smallest accepted violation fraction (0.1353), while the joint multi-output forest has 0.1419. This difference concerns the composition of the accepted set, not necessarily how many risky cases are missed in total, because the methods accept different fractions of the sample (0.4460 and 0.4738, respectively). Reading acceptance, violation, and recall together is therefore more informative than treating any one column as a complete measure of screening performance.

4.4 Budget-level stability

The mean over budget pairs hides variation in the evaluation target. Table 2 summarizes each estimator's lowest and highest mean F1 among the 21 eligible pairs, the standard deviation of its 21 pair means, and the number of pairs in which it ranks first. The final two columns describe sensitivity to the chosen budget grid and are not sampling uncertainty estimates. The table complements the five-repeat within-pair standard deviation in Table 1: one measures how scores move as the budget-defined classification task changes, while the other measures repeat variation for a fixed budget.

Weighted KNN has the smallest across-budget standard deviation (0.0174), followed by ExtraTrees (0.0200) and the multi-output random forest (0.0229). KNN ranks first in 12 of 21 pairs and the grouped MLP in six; the forest, RBF-SVR, and spline logistic each rank first once. Thus the best mean F1 estimator also wins most individual pairs, but its advantage does not extend to every budget. The MLP has the largest observed pair mean (0.8957), illustrating that local maxima and average performance answer different questions. The prevalence baseline is especially sensitive to the budget grid (standard deviation 0.3530), including pairs where its constant prediction yields no positive decisions and hence zero F1. This is a property of its fixed threshold rule, rather than evidence of a numerical failure.

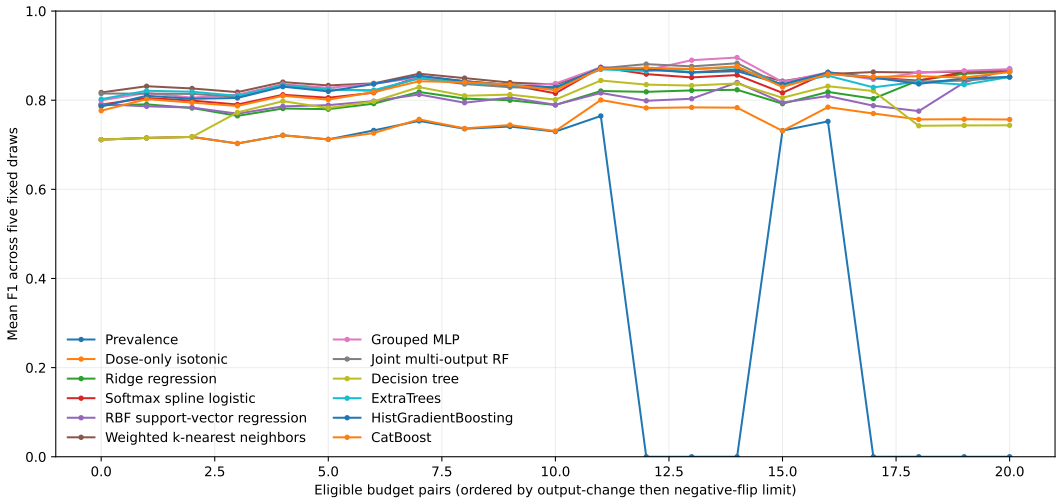


Figure 3: Sensitivity of mean balanced F1 to the 21 output-change/negative-flip budget pairs. Predictions remain frozen while the joint positive label changes with the budgets.

4.5 Budget sensitivity and applicability

Figure 3 shows F1 over all 21 budget pairs for each method. The winning model varies: weighted nearest neighbors has the highest F1 in 12 pairs, the grouped MLP in six, and three other estimators each lead one pair. The jointly fitted forest is the sole winner in one pair, while no constant or dose-only estimator leads. The forest and nearest-neighbor curves vary less across the tested budgets than the prevalence curve; the latter reaches zero F1 for some labels because its constant prediction produces no positive classifications at those operating points. Predictions remain frozen throughout this sweep, so the variation reflects the changing budget-defined labels rather than model refitting. The plot is a sensitivity analysis over evaluation criteria, not a hyperparameter selection procedure.

This budget dependence has a practical interpretation. A pair with a more permissive output-change limit can classify a different set of replacements as acceptable even when its negative-flip budget is unchanged. That changes both the positive examples and the decision rule applied to each model's frozen risk estimates. The models have different shapes and orderings of predicted risks, so their F1 curves need not move in parallel as the labels change. Averaging over the 21 pairs offers a compact summary, but it gives each selected pair equal influence and can obscure which risk tolerance a deployment owner values. A deployment decision should report the relevant budget-specific confusion matrix and review cost alongside the average comparison.

The transfer policy assigns 758 records to the exact-zero state, 1,308 records within role-specific dose support, 110 below support, and 563 above support. Every record receives a numeric prediction for balanced scoring, but its support status is retained. The 673 below- or above-support records therefore carry weaker dose support than in-range records.

4.6 Limitations

The study measures transfer within one packaged data snapshot, 13 models, seven task scopes, and 2,739 replacement records. The perturbation and replacement tables share 1,299 context identities. This evaluation therefore tests transfer across discrepancy-generating mechanisms within covered contexts and does not establish generalization to unseen models or contexts. Maximum absolute discrepancy also discards the sign, distribution, correlation, and accumulation pattern of errors across invocations. Upper-tail labels are sparse, and sampling with replacement can repeatedly select the same high-risk records. Support clipping defines predictions outside the fitted range but does not guarantee calibration for unseen discrepancies. The replacement labels come from the current packaged snapshot and are not identical to every historical validation snapshot reported in the source study.

Several design choices limit how broadly the comparison can be interpreted. The model settings are fixed rather than selected through nested, context-grouped validation, and computational budgets differ across linear, neighbor, neural, and ensemble estimators. This is an empirical comparison of these implementations and settings, not a search for the best achievable version of each family. The task-role taxonomy is shared across models, so its benefit is not independently measured here. The replacement pool also shares context identities with perturbation training; although replacement outcomes remain withheld from fitting, this is transfer across discrepancy-generating mechanisms within represented contexts rather than a holdout of unseen contexts. A stronger generalization test would reserve complete contexts or models before any estimator fitting and preserve that split across all model families.

The balanced protocol introduces a further boundary on claims. Equal class allocation supports comparison of ranking and threshold behavior when positive and negative examples receive equal evaluation weight. It does not estimate calibration, natural-prevalence precision, or expected deployment cost. Sampling records with replacement means the 105,000 draw positions are not 105,000 independent replacements; some records recur across repeats and budget pairs. The reported repeat standard deviations characterize only the chosen seed and resampling procedure. They omit uncertainty from candidate-seed selection in the source protocol, the replacement snapshot itself, and retraining on new perturbation data. Future work should evaluate frozen estimators on an independently collected replacement pool, report paired intervals over context-level resampling, and assess calibration at the natural event rate.

5 Conclusion

We compared twelve estimators using a shared balanced replacement evaluation. Their results vary by metric and budget pair: weighted nearest neighbors and a grouped MLP have the highest average F1, the softmax spline logistic model has the lowest accepted violation fraction, and the shallow tree has the strongest recall in the two severe accuracy-drop groups. No method leads consistently, and average-score differences among the leading estimators are small relative to repeat-level resampling variation. The jointly fitted random forest is one competitive point in this comparison, constructed from 120 depth-limited trees trained on dose, task, dtype, and operator role. These findings support reporting model trade-offs and the target operating budget together. Claims remain limited to contexts and discrepancy support represented in the current snapshot.

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References

[1] Nicholas J. Higham. *Accuracy and Stability of Numerical Algorithms*. Society for Industrial and Applied Mathematics, second edition, 2002. ISBN 978-0-89871-521-7. doi: 10.1137/1.9780898718027. URL https://doi.org/10.1137/1.9780898718027.

[2] Hashim Sharif, Prakalp Srivastava, Muhammad Huzaifa, Maria Kotsifakou, Keyur Joshi, Yasmin Sarita, Nathan Zhao, Vikram S. Adve, Sasa Misailovic, and Sarita Adve. ApproxHPVM: A portable compiler IR for accuracy-aware optimizations. *Proceedings of the ACM on Programming Languages*, 3(OOPSLA), 2019. doi: 10.1145/3360612. URL https://doi.org/10.1145/3360612.

[3] Chenhao Jiang, Anand Jayarajan, Hao Lu, and Gennady Pekhimenko. Arbitor: A numerically accurate hardware emulation tool for DNN accelerators. In *2023 USENIX Annual Technical Conference (USENIX ATC 23)*, pages 519–536, Boston, MA, 2023. USENIX Association. URL https://www.usenix.org/conference/atc23/presentation/jiang-chenhao.

[4] Jiayi Yuan, Hao Li, Xinheng Ding, Wenya Xie, Yu-Jhe Li, Wentian Zhao, Kun Wan, Jing Shi, Xia Hu, and Zirui Liu. Understanding and mitigating numerical sources of nondeterminism in LLM inference, 2025. URL https://arxiv.org/abs/2506.09501.

[5] Jay Shah, Ganesh Bikshandi, Ying Zhang, Vijay Thakkar, Pradeep Ramani, and Tri Dao. FlashAttention-3: Fast and accurate attention with asynchrony and low-precision. In *Advances in Neural Information Processing Systems*, volume 37, 2024. URL https://arxiv.org/abs/2407.08608.

[6] Anne Ouyang, Simon Guo, Simran Arora, Alex L. Zhang, William Hu, Christopher Ré, and Azalia Mirhoseini. KernelBench: Can LLMs write efficient GPU kernels?, 2025. URL https://arxiv.org/abs/2502.10517.

[7] Jianling Li, Shangzhan Li, Zhenye Gao, Qi Shi, Yuxuan Li, Zefan Wang, Jiacheng Huang, Haojie Wang, Jianrong Wang, Xu Han, Zhiyuan Liu, and Maosong Sun. TritonBench: Benchmarking large language model capabilities for generating Triton operators, 2025. URL https://arxiv.org/abs/2502.14752.

[8] Anonymous Author(s). Understanding Tolerance Threshold Setup in Automated GPU Kernel Optimization. Unpublished manuscript used as the reference protocol for this course project, 2026.

[9] Zhongzhen Wen, Yinghui Zhang, Zhong Li, Zhongxin Liu, Linna Xie, and Tian Zhang. MultiKernelBench: A multi-platform benchmark for kernel generation, 2025. URL https://arxiv.org/abs/2507.17773.

[10] Alexander Henzi, Johanna F. Ziegel, and Tilmann Gneiting. Isotonic distributional regression. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 83(5):963–993, 2021. doi: 10.1111/rssb.12450. URL https://doi.org/10.1111/rssb.12450.

[11] Leo Breiman. Random forests. *Machine Learning*, 45(1):5–32, 2001. doi: 10.1023/A:1010933404324. URL https://doi.org/10.1023/A:1010933404324.

[12] Pierre Geurts, Damien Ernst, and Louis Wehenkel. Extremely randomized trees. *Machine Learning*, 63(1):3–42, 2006. doi: 10.1007/s10994-006-6226-1. URL https://doi.org/10.1007/s10994-006-6226-1.

[13] Anna Veronika Dorogush, Vasily Ershov, and Andrey Gulin. CatBoost: Gradient boosting with categorical features support, 2018. URL https://arxiv.org/abs/1810.11363.

[14] Sijie Yan, Yuanjun Xiong, Kaustav Kundu, Shuo Yang, Siqi Deng, Meng Wang, Wei Xia, and Stefano Soatto. Positive-congruent training: Towards regression-free model updates. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2021. URL https://arxiv.org/abs/2011.09161.

[15] OpenAI. Codex. Software tool, 2025. URL https://github.com/openai/codex.